

Transport of Water and Solutes across Maize Roots Modified by Puncturing the Endodermis¹

Further Evidence for the Composite Transport Model of the Root

Ernst Steudle*, Martina Murrmann, and Carol A. Peterson

Lehrstuhl für Pflanzenökologie, Universität Bayreuth, D-95440 Bayreuth, Germany (E.S., M.M.); and Department of Biology, University of Waterloo, Waterloo, Ontario, Canada N2L 3G1 (C.A.P.)

The effects of puncturing the endodermis of young maize roots (*Zea mays* L.) on their transport properties were measured using the root pressure probe. Small holes with a diameter of 18 to 60 μm were created 70 to 90 mm from the tips of the roots by pushing fine glass tubes radially into them. Such wounds injured about 10^{-2} to $10^{-3}\%$ of the total surface area of the endodermis, which, in these hydroponically grown roots, had developed a Casparian band but no suberin lamellae. The small injury to the endodermis caused the original root pressure, which varied from 0.08 to 0.19 MPa, to decrease rapidly (half-time = 10–100 s) and substantially to a new steady-state value between 0.02 and 0.07 MPa. The radial hydraulic conductivity (L_p) of control (uninjured) roots determined using hydrostatic pressure gradients as driving forces was larger by a factor of 10 than that determined using osmotic gradients (averages: L_p [hydrostatic] = $2.7 \times 10^{-7} \text{ m s}^{-1} \text{ MPa}^{-1}$; L_p [osmotic] = $2.2 \times 10^{-8} \text{ m s}^{-1} \text{ MPa}^{-1}$; osmotic solute: NaCl). Puncturing the endodermis did not result in measurable increases in hydraulic conductivities measured by either method. Thus, the endodermis was not rate-limiting root L_p ; apparently the hydraulic resistance of roots was more evenly distributed over the entire root tissue. However, puncturing the endodermis did substantially change the reflection (σ_{sr}) and permeability (P_{sr}) coefficients of roots for NaCl, indicating that the endodermis represented a considerable barrier to the flow of nutrient ions. Values of σ_{sr} decreased from 0.64 to 0.41 (average) and P_{sr} increased by a factor of 2.6, i.e. from 3.8×10^{-9} to $10.1 \times 10^{-9} \text{ m s}^{-1}$ (average). The roots recovered from puncturing after a time and regained root pressure. Measurable increases in root pressure became apparent as soon as 0.5 to 1 h after puncturing, and original or higher root pressures were attained 1.5 to 20 h after injury. However, after recovery roots often did not maintain a stable root pressure, and no further osmotic experiments could be performed with them. The Casparian band of the endodermis is discontinuous at the root tip, where the endodermis has not yet matured, and at sites of developing lateral roots. Measurements of the cross-sectional area of the apoplastic bypass at the root tip yielded an area of 0.031% of the total surface area of the endodermis. An additional 0.049% was associated with lateral root primordia. These areas are larger than the artificial bypasses created by wounding in this study and may provide pathways for a “natural bypass flow” of water and solutes across

the intact root. If there were such a pathway, either in these areas or across the Casparian band itself, roots would have to be treated as a system composed of two parallel pathways (a cell-to-cell and an apoplastic path). It is demonstrated that this “composite transport model of the root” allows integration of several transport properties of roots that are otherwise difficult to understand, namely (a) the differences between osmotic and hydrostatic water flow, (b) the dependence of root hydraulic resistance on the driving force or water flow across the root, and (c) low reflection coefficients of roots.

There is increasing evidence that pathways for the radial transport of water across roots differ for different species. In some roots, a cell-to-cell transport of water predominates, whereas in others water flow is mainly around cells, i.e. apoplastic (Steudle, 1989, 1992, 1993a, 1993b). Examples of roots with mostly cell-to-cell transport are young barley and bean, whereas in young maize and cotton roots, an apoplastic transport dominates (Steudle and Jeschke, 1983; Steudle et al., 1987; Radin and Matthews, 1989; Steudle and Brinckmann, 1989). In species exhibiting large differences between osmotic and hydrostatic water flow, the extent to which apoplastic transport occurs depends on the nature of the force set up between root medium and xylem to drive water flow. Hydrostatic gradients (which dominate in the intact, transpiring plant) cause a much larger flow due to a higher L_p than osmotic, and the differences have been found to be as large as an order of magnitude or even more (Steudle and Frensch, 1989; Zhu and Steudle, 1991; Cruz et al., 1992; Birner and Steudle, 1993). Differences between species were not due to different contributions of the longitudinal (axial)

Abbreviations: A_r , root surface area (m^2); D_s , diffusion coefficient ($\text{m}^2 \text{ s}^{-1}$); ϵ_x , elastic coefficient of root xylem (MPa); J_{vr} , radial water flow across root (m s^{-1}); J_{sr} , radial solute flow across root ($\text{mol m}^{-2} \text{ s}^{-1}$); J_{sr}^* , active component of radial solute flow ($\text{mol m}^{-2} \text{ s}^{-1}$); k_{sr} (k_{sr}), rate constant of water (solute) exchange across root (s^{-1}); L_p , radial hydraulic conductivity of root ($\text{m s}^{-1} \text{ MPa}^{-1}$); P_E , final root pressure (MPa); P_r , root pressure; $P_{\min}$ ($P_{\max}$), minimum (maximum) root pressure; P_{tor} , steady-state root pressure; P_{sr} , permeability coefficient of root (m s^{-1}); $T_{1/2}^w$, half-time of water exchange across root (s); $T_{1/2}^s$, half-time of solute exchange across root (s); V_x , volume of mature root xylem (m^3); $\Delta P_r/\Delta V_s$, elastic coefficient of root pressure probe (MPa m^{-3}); σ_{sr} , reflection coefficient of root (1).

¹ Supported by a grant from EUROSILVA (project No. 39473C) to E.S. and by a Bilateral Exchange Grant jointly funded by the Deutsche Forschungsgemeinschaft and the Natural Sciences and Engineering Research Council of Canada to C.A.P.

* Corresponding author; fax 49–921–55–2564.

hydraulic resistances because, in all cases, the longitudinal resistance was negligible compared with the radial (Frensch and Steudle, 1989). Variations in L_p coincide with relatively low σ_{sr} for solutes for which cell membranes would exhibit a reflection coefficient of virtual unity (salts, sugars, etc.).

The differences in osmotic and hydrostatic water flow and the low σ_{sr} could not be explained by effects of unstirred layers (Steudle and Frensch, 1989). Instead, a new model was established by adapting basic theories of transport across composite membranes to the complex osmotic barrier of roots (Steudle, 1989, 1992, 1993a, 1993b). In the model the root is considered to be a composite of concentric layers of different tissues in series, i.e. the epidermis, cortex, and stele, and two pathways in parallel, i.e. the apoplast and the cell-to-cell path. These elements represent a network of transport resistances that can be analyzed using basic principles of the thermodynamics of irreversible processes (Kedem and Katchalsky, 1963; House, 1974), whereby parallel elements (cell-to-cell and apoplastic paths) appear to be the most pertinent (Steudle, 1989, 1992, 1993a, 1993b).

The composite transport model of the root explains, in principle, low σ_{sr} and differences between osmotic and hydrostatic water flow. It also resolves the old but still unsolved problem of variable hydraulic resistance of roots, i.e. the increase of the hydraulic conductivity with increasing water flow across roots (Fiscus, 1975; Weatherley, 1982; Passioura, 1988; Steudle, 1989). In the model the existence of some water flow through the apoplast bypassing the symplast is crucial. This focuses attention on discontinuities naturally present in the endodermis that could form the structural basis of apoplastic bypasses for water. In young roots such discontinuities are at the root tip where the endodermis is immature and at lateral root primordia. Alternatively, the mature Casparian band itself may not be completely impermeable to water and solutes. With respect to lateral root initials, Karas and McCully (1973; p. 266) stated that "...there was no indication of Casparian strips or suberized lamellae in the oldest laterals examined along their complete length (those emerged by about 4.5 mm; Bell and McCully, 1970). Thus, lateral primordia present a relatively large surface of unsuberized cell walls to the contents of the apoplast of the cortex, and for some time after emergence, to the outer environment."

On the other hand, the young cell walls, both in the laterals and at the root tip, are less permeable to apoplastic dyes than mature primary walls (Peterson et al., 1981; Enstone and Peterson, 1992) and, therefore, these walls may also represent a relatively high hydraulic resistance. There are virtually no data concerning the transport of water and solutes across the Casparian band. However, comparisons between the hydrostatic hydraulic conductivity of roots and that of root cells using pressure probes (root and cell level) and other techniques indicate some apoplastic bypass of water either on routes lacking a Casparian band or across the endodermis (Steudle et al., 1987; Radin and Matthews, 1989; Zhu and Steudle, 1991; Steudle and Frensch, 1989; Birner and Steudle, 1993).

In addition to the anatomical work and the transport studies using the root pressure probe, there is also evidence for small apoplastic bypasses from measurements of solute

uptake into roots. For example, Hanson et al. (1985) detected entry of the apoplastic dye trisodium-8-hydroxy-1,3,6-pyrenetrisulfonate into roots of red pine. In rice Yeo et al. (1987) reported an apoplastic bypass flow of both NaCl and trisodium-8-hydroxy-1,3,6-pyrenetrisulfonate across the roots. Recently, Frensch et al. (1992) found, by x-ray microanalysis of cross-sections of young maize roots treated with NaCl, that this solute appeared more rapidly in the root xylem than in cortical root cells, which supported the view of a bypass of root protoplasts.

In this investigation we have explored the role of apoplastic bypasses during water and solute transport in young maize (*Zea mays* L.) roots by creating small holes extending through the cortex including the endodermis, and sometimes part of the stele, thus producing additional bypasses. The holes were made with the tip of a microcapillary tube, 18 to 60 μm in diameter. The effects of this injury on root pressure, as well as on the hydraulic conductivity and σ_{sr} and P_{sr} , were measured using the root pressure probe. The changes caused by small artificial bypasses in the endodermis were compared with those naturally existing in roots as described above. The comparison should help to clarify the role of apoplastic transport components in roots as long as direct measurements of water and solute flows along the cell-to-cell and apoplastic pathways cannot be made for technical reasons.

MATERIALS AND METHODS

Plant Material

Karyopses of maize (*Zea mays* L. cv Tanker; Südwestdeutsche Saatzucht, Rastatt, Germany) were germinated for 2 to 4 d in the dark on filter paper soaked with 0.1 mM CaCl_2 and were then grown in continuous immersion hydroponics under a 14-/10-h light/dark regime at 20/17°C and a light intensity of 420 $\mu\text{E m}^{-2} \text{s}^{-1}$ (Steudle and Frensch, 1989; Peterson and Steudle, 1993). The aerated culture solution contained the following macronutrients (in mM): 0.60 K^+ , 0.30 Ca^{2+} , 0.40 Mg^{2+} , 0.80 NH_4^+ , 1.60 NO_3^- , 0.30 PO_4^{3-} , and 0.30 SO_4^{2-} and micronutrients (in μM): 50 Cl^- , 36 Fe^{3+} , 36 Na^+ , 25 BO_3^{3-} , 2 Mn^{2+} , 2 Zn^{2+} , 0.5 Cu^{2+} , and 0.5 MoO_4^{2-} at a pH of 6.0 adjusted with H_2SO_4 . Fe^{3+} and Na^+ were supplied as FeNa-EDTA. Plants were 7 to 14 d old when used in experiments.

Anatomy

The area of the endodermal cylinder was calculated from measurements of its diameter in root cross-sections and the length of the root minus the hydraulically inert tip (21.5 mm; Peterson and Steudle, 1993). Thus, the effective length of the endodermal cylinder was used instead of the geometric length, because the former would be relevant in transport studies. Estimates of specific apoplastic areas were made from various sections in which the walls were stained with the fluorescent dye Cellufluor (0.01%, w/w). To estimate the apoplastic area of the endodermis at the level of the Casparian band, lengths and tangential widths of endodermal cells were measured from freehand, longitudinal sections, and cross-sections of typical roots. The thickness of the cell walls was 1 μm . Total and apoplastic areas were also cal-

culated for regions lacking a Casparian band, i.e. for root tips and for lateral root primordia. The total area available at root tips was evaluated from the diameter of the endodermal cylinder 4 mm from the root tip, the point where the Casparian band was just maturing. The apoplasmic area was estimated from cross-sections stained with Cellufluor. Similarly, the total area of the endodermis interrupted by a lateral root primordium was calculated from freehand cross- and longitudinal sections of the main root, which were median longitudinal through the lateral. To estimate the fractional apoplasmic area of this region, cell diameters of the lateral root primordia were measured at the level of the main root endodermis. The type of section that proved most useful for this purpose was a paradermal, longitudinal section at the level of the endodermis of the main root positioned to provide a cross-section of the lateral. For these preparations it was necessary to use laterals that had emerged from the parent roots to orient them for sectioning. Data for the cell wall thickness were taken from electron micrographs reported by Karas and McCully (1973). The total number of unemerged lateral root primordia, plus those emerged by 2 mm, were counted on five roots that had been cleared and stained using the method of Hackett and Stewart (1969), i.e. by staining the nuclear material (which is very dense in the mitotically active primordia) with acetocarmine and clearing most of the remaining cytoplasmic substances with methyl benzoate. This gave a conservative estimate of the numbers of laterals without Casparian bands because, according to Karas and McCully (1973), laterals of maize that had emerged by as much as 4.5 mm still had no Casparian bands. Sections and stained

roots were photographed with slide film, white-light illumination with Ektachrome 64T (ISO 64), and UV/violet illumination (excitation wavelengths: 390–420 nm) with Fujichrome (ISO 100).

P_r and Lp_r

Roots were excised under water near the kernel to obtain an unbranched segment of primary root 89 to 174 mm long (diameter: 0.78–1.28 mm). To prepare for an experiment, a root was inserted into a silicone seal (8–10 mm long) that was then placed in the root pressure probe (Fig. 1). The seal was carefully tightened stepwise until the root pressure began to increase after tightening. This procedure ensured that the seal around the root was sufficiently closed to prevent leakage of liquid through the root cortex. Whether or not the xylem vessels in the sealed area remained open or were crushed was tested at the conclusion of each experiment as described below (Steudle and Frensch, 1989; Peterson and Steudle, 1993). Steady-state root pressures between 0.08 and 0.19 MPa (0.8–1.9 bars) were obtained after 1 to 2 h, and experiments commenced. The Lp_r was evaluated from root pressure relaxations.

Water flows were induced by either changing the root pressure with the aid of a movable metal rod in the equipment (Fig. 1; hydrostatic experiments) or by altering the osmotic pressure of the medium. The latter was accomplished by changing the bathing solution around the root from the normal culture solution to one amended with a specific amount of NaCl as an osmoticum (exosmotic experiment). In

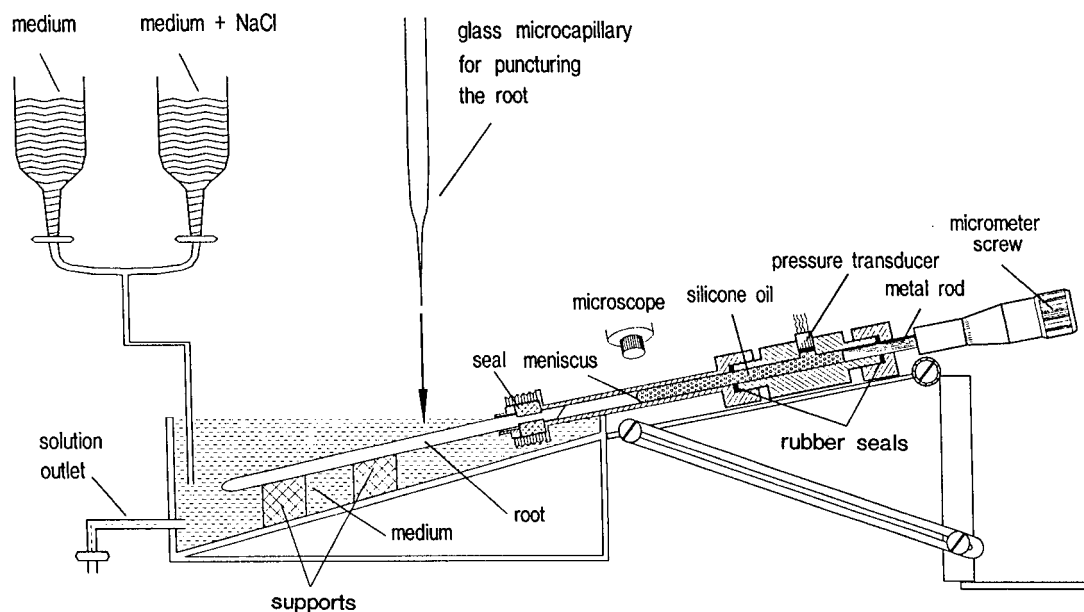


Figure 1. Diagram of a root pressure probe with a root fixed to it. To induce water flows (root pressure relaxations), the stationary root pressure could be changed either by moving the metal rod (hydrostatic experiments) or by exchanging the root-bathing medium for one with a different osmotic concentration (osmotic experiments). Root pressure was measured continuously with a pressure transducer. From the rate constants (half-times) of the root pressure relaxations measured, the Lp_r , σ_{sr} , and P_{sr} were calculated. To puncture the root endodermis, the tip of a thin microcapillary tube was inserted radially into the root with the aid of a micromanipulator (not shown). For further explanation, see text.

endosmotic experiments, the change was made back to the original culture solution. The changes of root pressure, recorded after step changes in water potential, were exponential with time. From the rate constant (k_{wr}) of the processes, the Lp_r was evaluated (Steudle et al., 1987; Steudle, 1989, 1992, 1993a, 1993b):

$$k_{wr} = \frac{\ln(2)}{T_{1/2}^w} = A_r \frac{\Delta P_r}{\Delta V_s} Lp_r. \quad (1)$$

$T_{1/2}^w$ is the half-time of the process; $\Delta P_r/\Delta V_s$ (in MPa m⁻³) is the elastic coefficient of the measuring system (the inverse of its water capacity; V_s = water volume of system), and A_r is the effective surface area of the root. $\Delta P_r/\Delta V_s$ was determined by inducing step changes of volume (ΔV_s) with the aid of the movable rod and measuring the resulting changes of root pressure (ΔP_r ; Steudle and Jeschke, 1983). The effective surface area of the root was calculated from its diameter and length, subtracting the apical part where tracheary elements of the early and late metaxylem were immature, i.e. the terminal 21.5 mm (Peterson and Steudle, 1993).

In osmotic experiments with permeating osmotica (as in this paper), root pressure relaxations were biphasic, and the phases were separated by minima or maxima of root pressure, i.e. water flow across the root changed direction during the experiment. The first rapid phase ("water phase") was caused by the change of the osmotic pressure of the medium. The second, slower phase was due to the passive permeation of the solute across the root cylinder, which changed the osmotic gradient across the root causing water to enter ("solute phase"). Both phases could be fitted by a single exponential curve ($r^2 \geq 0.98$).

In hydrostatic experiments, pressure relaxations caused by a change of root pressure exhibited only one phase. This was either an uptake of water by the root (endosmotic water flow) or an extrusion of water from the root (exosmotic water flow), depending on whether the pressure in the head of the probe was decreased or increased. Hydrostatic relaxation curves were composed of three distinct regions brought about by different rates of changes of root pressure with time, i.e. three different phases could be distinguished. There was an initial very rapid phase immediately following the change in P_r (half-time: 0.7–0.9 s), then a central body of the curve (half-time: 1.9–4.9 s), and finally a very slow component (half-time: 52–66 s). As discussed elsewhere (Peterson and Steudle, 1993), the second phase is related to the radial exchange of water across the root cylinder. The two other components could easily be separated from this phase because of the large differences in their rate constants. To calculate the rates of radial water exchange across the root, chart recorder strips were digitized on a graphics tablet, and the curves were then analyzed using fit programs. The programs also assisted in finding the final root pressure (P_E ; see Fig. 3) of pressure relaxations. The use of a proper value of P_E had some impact on the evaluation of k_{wr} ($T_{1/2}^w$) (Birner and Steudle, 1993; Melchior and Steudle, 1993).

In a typical experiment in which the hydraulic conductivity of a root was measured, the $\Delta P_r/\Delta V_s$ was determined initially, followed by six replicate hydrostatic relaxations and one osmotic pressure relaxation. The latter biphasic relaxations

also gave P_{sr} and σ_{sr} (see below) of the root to NaCl. After these measurements the root endodermis was punctured with the aid of a glass tube that had a tip diameter of 18 to 60 μm (Fig. 1). It was prepared by drawing a molten capillary tube on a vertical puller used to prepare microelectrodes. The microcapillary tube was advanced slowly, radially into the root, and then withdrawn. In this way, a single hole or a few holes were produced in the cortex, endodermis, and sometimes in part of the stele; in all cases, mature xylem vessels were not damaged. Holes were made at distances between 70 and 90 mm from the root tip. Puncturing led to a reduced steady root pressure (P_{ro}). After puncturing all procedures were repeated to measure the transport coefficients Lp_r , σ_{sr} , and P_{sr} . At the conclusion of each experiment, roots were cut with a razor blade right at the seal, which resulted in an immediate decrease of pressure to zero (atmospheric). Additional hydrostatic pressure relaxations were performed, and these were usually very fast, indicating that, within the sealing area, the vessels remained open and that the hydraulic resistance of the xylem within the seal was very low compared with that of the intact root during the experiments (which lasted for 4–10 h for each root). Experiments were disregarded when the half-times of pressure relaxations after severing the root were larger than one-fifth to one-tenth of that obtained for the intact system.

Measurement of P_{sr} and σ_{sr} Coefficients

Using a permeating solute (NaCl), we calculated P_{sr} from the rate constant (k_{sr}) of the second phase (solute phase) of biphasic osmotic pressure relaxations (Steudle, 1989, 1992, 1993a), i.e.:

$$k_{sr} = \frac{\ln(2)}{T_{1/2}^s} = \frac{A_r}{V_x} P_{sr} \quad (2)$$

where $T_{1/2}^s$ indicates half-time of the solute phase, and V_x is the volume of mature root xylem. For the roots used, V_x was 3% of the total root volume as estimated from root cross-sections obtained at different distances from the root tip (Steudle and Frensch, 1989). Thus, after measuring k_{sr} ($T_{1/2}^s$), V_x , and A_r (see above), we calculated P_{sr} from Equation 2.

σ_{sr} for the given solute was obtained from the maximum change of root pressure (ΔP_r) caused by a change of the osmotic pressure of the medium. During hypertonic changes $\Delta P_r = P_{ro} - P_{rmin}$ and during hypotonic changes $\Delta P_r = P_{ro} - P_{rmax}$ [P_{ro} = steady-state root pressure; $P_{rmin(max)}$ = minimum (maximum) root pressure]. Reflection coefficients were calculated from Steudle et al. (1987) and Steudle (1989, 1992, 1993a, 1993b):

$$\sigma_{sr} = \frac{\Delta P_r}{\Delta \pi_s} \frac{\epsilon_x + \pi_x}{\epsilon_x} \exp(k_{sr} \cdot t_{min}), \quad (3)$$

where $\Delta \pi_s$ is the change of the osmotic pressure in the medium of permeating solute "s" (NaCl), ϵ_x is the elastic modulus of the root xylem, and π_x is the osmotic pressure in the root xylem. Because the elastic modulus of the root xylem was large, $\epsilon_x \gg \pi_x$ was valid, and the second term on the right side of Equation 3 was equal to unity to a good approx-

imation. The third factor on the right side of the equation ($\exp[k_{sr} \cdot t_{\min}]$) corrected for the solute flow into or out of the xylem during the water phase, i.e. during the time taken to reach the minimum or maximum ($t_{\min}$). As for the osmotic and hydrostatic Lp_r , the coefficients P_{sr} and σ_{sr} were measured before and after puncturing the endodermis.

RESULTS

Anatomy

The endodermis of a typical maize root used in this work can be considered a cylinder with a diameter of 0.50 mm and a length of 85 mm, i.e. the total length of the root less the hydraulically isolated tip region of 21.5 mm. The total surface area of such an endodermis is 134 mm². The average length and tangential width of its component cells were 280 and 18 μm , respectively. Assuming a wall thickness of 1 μm (see "Materials and Methods"), the per cell area of the apoplast in a tangential longitudinal view, sectioning midway through the endodermis, would be 596 μm^2 . Thus, the total area of the apoplast in this plane of the entire endodermis would be 16 mm² or 12% of the total surface area. Endodermal cells possessed Casparian bands but not suberin lamellae (Fig. 2A). The exodermis was immature, i.e. with neither Casparian bands nor suberin lamellae (Fig. 2A).

A discontinuity in the Casparian band exists at the root tip where the structure is immature. The diameter of the endodermis 4 mm from the root tip was $2R = 350 \mu\text{m}$. The cells of the stele were small in diameter ($2r = 9.5 \mu\text{m}$) and closely packed (Fig. 2B). Assuming a wall thickness (Δr) of 1 μm , the proportion of one cell occupied by wall would be $2\pi r \cdot \Delta r / \pi r^2 = 2\Delta r / r = 2 \cdot 1 / 9.5 = 0.21$. Therefore, the wall area in the whole apical bypass would be $\pi R^2 \cdot 0.21 = \pi \cdot 175^2 \cdot 0.21 = 40,400 \mu\text{m}^2$, i.e. 0.030% of the endodermal surface area. Because the endodermis matures earlier than the tracheary elements, water and solutes following an apoplastic path would have to cross the epidermis and cortex and then move basipetally across the young stelar tissue before reaching a mature vessel. Although this is not a long path, the permeability of these walls is low compared to older walls, at least for the apoplastic dye berberine (Enstone and Peterson, 1992). This has to be taken into account when considering both the hydraulic and diffusional resistances of the tip route and its possible contribution to the overall apoplastic bypass.

Discontinuities in the Casparian band are associated with lateral root primordia, which are very numerous in maize (Fig. 2C). On average, the roots possessed 170 young laterals, which were within the main root or had emerged 2 mm from its surface. Their average diameters at the level of the main root endodermis were determined from measurements of cross-sections and longitudinal sections of the main root, which were median longitudinal sections of the primordia (Fig. 2D). The discontinuous area of the endodermis was usually less than the maximum diameters of the laterals, because primordia were narrower at the level of the endodermis than in the nearby cortex (Fig. 2D). The total circular area in which the Casparian band of the main root was absent at the base of a primordium was 30,000 μm^2 . An estimate of the potential apoplastic bypass was made from

paradermal, longitudinal sections taken at the level of the endodermis (Fig. 2E). It was possible to recognize an appropriate section because the cells of the main root endodermis possessed large primary pit fields that were not present in pericycle or neighboring cortical cells (Fig. 2E). The mean diameter of the cells in the initials was $2r = 11 \mu\text{m}$ and the mean cell wall thickness $\Delta r = 0.035 \mu\text{m}$ (0.025–0.050 μm ; Karas and McCully, 1973). This resulted in a fraction available for apoplastic transport of $2\pi r \cdot \Delta r / \pi r^2 = 2\Delta r / r = 2 \cdot 0.035 / 11 = 0.0064$ or 0.64% of the total cross-section. Because the cross-section of a single root primordium was, on average, 30,000 μm^2 (mean diameter: 195 μm), the absolute value of apoplast available in a lateral was 390 μm^2 . There were 170 initials in a root, giving a total apoplastic cross-section of 66,300 μm^2 . This is only a small fraction, i.e. 0.049% of the total surface area of the endodermis. By contrast, the cross-section of the apoplastic route across the Casparian band of the endodermis was as large as 12% of the total surface area of the endodermis (see above), i.e. it was larger by a factor of 245 than that available in the apoplast of the initials.

Root Pressure and Hydraulic Measurements

In a typical example, root pressure, osmotic and hydrostatic hydraulic conductivities, and reflection coefficients before and after puncturing the endodermis were all measured on one root (Fig. 3). In Figure 3A the initial four pressure spikes represent measurements of the elastic coefficient of the system ($\Delta P_r / \Delta V_s = 2.9 \times 10^9 \text{ MPa m}^{-3}$ or 29 bar μL^{-1} for this experiment). These measurements of elasticity were followed by hydrostatic (Fig. 3, B and D) and osmotic (Fig. 3, C and E) pressure relaxations, including the solute phases for NaCl. From the half-times of hydrostatic pressure relaxations (Table I) and from the water phases of the osmotic experiments, Lp_r was calculated (Table III). From the solute phases of osmotic experiments, permeability and reflection coefficients were determined (Table II, see below). The hydrostatic hydraulic conductivity was considerably larger, and the $T_{1/2}^w$ values were considerably smaller than the osmotic (factor of 10, on average; Table III).

At the time indicated by an arrow in Figure 3F, a thin microcapillary tube (tip diameter: 18 μm) was slowly driven into the root through the cortex, endodermis, and part of the stelar tissue and was then withdrawn. This resulted in a direct injury to 2 to 10 endodermal cells that represent 10^{-2} to $10^{-3}\%$ of the total surface area of the endodermis (Peterson et al., 1993). Injuring the endodermis brought about a rapid decrease in root pressure (half-time = 100 s) due to an outflow of solution from the stele across the leak. A new, steady value of root pressure of about 0.04 MPa (0.4 bar) was obtained when the rate of active uptake of solutes (nutrients) compensated for their loss. The osmotic concentration of the solution in the apoplast of the stele apparently remained somewhat higher than that in the medium, causing a root pressure larger than atmospheric. It should be noted that the absolute value of the decrease in pressure was fairly variable in different experiments (see Tables I, III, and IV), which could have reflected variable damage to the endodermis.

When root pressure attained a new steady value after the

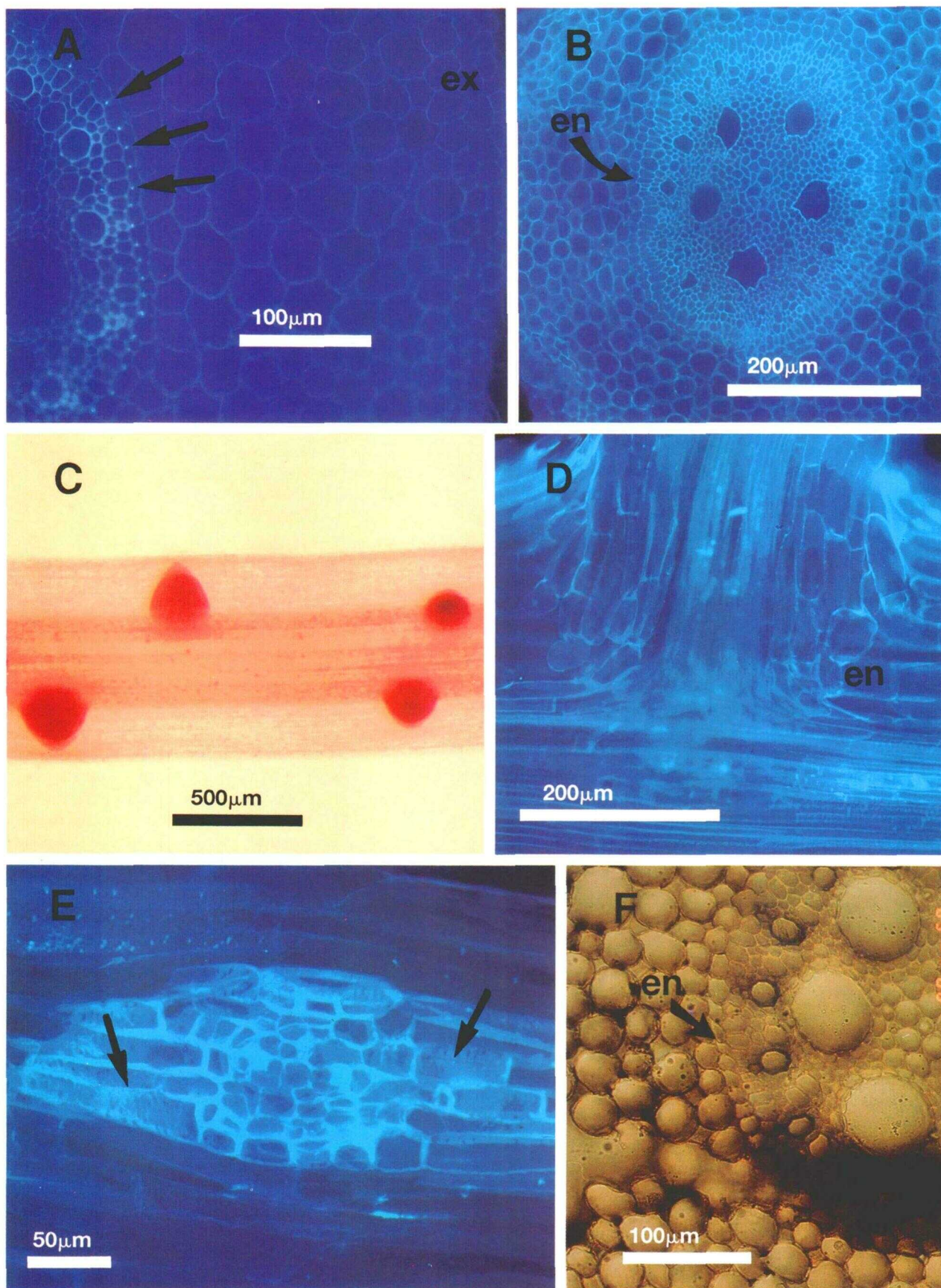


Figure 2. Photomicrographs of young maize roots typical of those used in the experiments. See text for details of the staining procedures. All pictures with the exception of C and F were obtained with epifluorescence. A, Transverse section of a root stained with berberine and counterstained with aniline blue (Brundrett et al., 1988). Casparian bands (arrows)

endodermis was punctured, the σ_{sr} and P_{sr} were remeasured for the modified root (Fig. 3, G–J; Tables I–III). It is evident from the data given in Figure 3 and in Tables I to III that hydraulic parameters (hydrostatic and osmotic $T_{1/2}^w$ and Lp_r) were not affected by puncturing. For example, the ratio of hydrostatic $T_{1/2}^w$ values measured before and after puncturing was 1.2 ± 0.3 (mean $\pm$ SD; Table I). Similarly, hydraulic conductivity did not change in osmotic experiments (Table III). The lack of an effect of puncturing is remarkable because of the common view that the endodermis should represent a main hydraulic barrier in the root. If this view were correct, puncturing this barrier should have resulted in a considerable shortening of half-times and in an increase of hydraulic conductivities.

In contrast to the hydraulic parameters, solute relations parameters (P_{sr} and σ_{sr}) did change upon puncturing the roots (Tables II and III). Control values of σ_{sr} and P_{sr} were similar to data already published (Azaizeh and Steudle, 1991; Steudle, 1992, 1993a, 1993b), that is, on average, $\sigma_{sr} = 0.64$ and $P_{sr} = 3.8 \times 10^{-9}$ m s $^{-1}$. This means that intact roots did not behave like ideal osmometers ($\sigma_{sr} = 1$ and $P_{sr} = 0$). Injuring only 10^{-2} to $10^{-3}\%$ of the total surface area of the endodermis resulted in a substantial (36%) reduction of σ_{sr} (on average, σ_{sr} was reduced from 0.64 to 0.41) and in an increase of P_{sr} by a factor of 2.7 on average (Tables II and III).

At the end of each experiment, the root was excised next to the silicone seal (arrow, Fig. 3K), and the root pressure immediately decreased to zero. Subsequent hydrostatic relaxations had a very short half-time, confirming that the vessels of the xylem had been open during the experiment (Fig. 3K). Previous work in which a Cellufluor solution was forced through the root while it was still in the seal also confirmed that the solution was flowing through the vessels in experiments of this type (Peterson and Steudle, 1993).

In one set of experiments, more than one hole was made in the root cylinder. The root pressure decreased stepwise after each endodermal puncture, reaching steady values each time (Fig. 4). Such experiments were a challenge to perform because they required more than one puncture of the endodermis. It was essential not to injure the root too deeply and thereby damage a mature vessel because this resulted in an irreversible decline of root pressure to zero. During prolonged experiments (longer than a few hours) with injured roots, root pressures tended to become unstable. Because of the length of time required for the multiple punctures and subsequent stabilizations of root pressure, it was possible to perform only the rapid hydrostatic measurements that yielded values for $T_{1/2}^w$ and Lp_r . The more lengthy osmotic

experiments that would have also yielded values of P_{sr} and σ_{sr} could not be done.

Recovery of Roots from Puncturing

Evidently, punctured roots were capable of regaining their root pressure (Fig. 5, Table IV). The time required for recovery was quite variable, ranging from 1.5 to 20 h to reach the original root pressure. In some cases, root pressures even higher than the original were achieved (see root numbers 13–15 and 18–20 in Table IV). As in the multiple puncturing experiments described above, only hydrostatic experiments ($T_{1/2}^w$, Lp_r) could be made. The results indicated that there was no change in hydrostatic Lp_r after recovery (Table IV). For some roots, cross-sections were taken through the punctured zones at the end of the experiments. Conspicuous browning of the walls of the endodermis and nearby cells of the stele was evident in unstained preparations (Fig. 2F).

DISCUSSION

Puncturing young maize roots affected transport of water and solutes quite differently. Steady-state root pressures were considerably reduced by the small injuries inflicted on the endodermis, but the $T_{1/2}^w$ and the Lp_r remained unchanged. The solute permeability, measured for NaCl as the test solute, increased by a factor of 3 and σ_{sr} decreased by one-third. The areas of artificial apoplastic bypasses created by puncturing were only 10^{-2} to $10^{-3}\%$ of the total surface area of the endodermis and of an order of magnitude or even smaller than those naturally present in intact roots in secondary root initials ($3 \times 10^{-2}\%$) or across the tip route ($5 \times 10^{-2}\%$). Thus, modified roots represent model systems to study the role of apoplastic components of water and solute transport across roots and their effect on root osmotic properties. They should be valuable systems for testing root transport models.

Steady-State Root Pressure of Modified Roots

Puncturing the endodermis caused a rapid decrease of root pressure with a half-time of 10 to 100 s. This half-time was substantially larger than that for water exchange in hydrostatic experiments but was smaller by 1 to 2 orders of magnitude than that measured for the diffusional flow of solutes after the root was punctured. Therefore, it is safe to conclude that the half-time was not rate limited by the diffusional flow of solutes (nutrients) from the xylem into the medium across the artificial pore or injury. Rather, it was caused by a convective flow of xylem sap across the hole. The efflux of

(Continued from facing page) are present in the endodermis but not in the exodermis (ex). B, Transverse section 4 mm from the tip stained with Cellufluor. The diameters of the immature endodermis (en) and those of the enclosed cells were used to estimate the area of the apoplastic bypass at the root tip. C, Whole mount of a cleared root stained with acetocarmine. Unemerged lateral roots were evident as dark red conical structures. D, Longitudinal section of a main root with a near-median longitudinal section of a lateral root, stained with Cellufluor. The diameter of the lateral root at its widest point is greater than the discontinuity in the main root endodermis (en). E, Longitudinal (paradermal) section of a main root with a cross-section of the base of a lateral at the level of the endodermis of the main root, stained with Cellufluor. The endodermis (arrows) is distinguished by the presence of large primary pit fields in its walls. Such sections were used to estimate the apoplastic bypass area associated with lateral root primordia. F, Transverse section of a root that had been punctured 24 h earlier. The outer walls of several cells of the endodermis (en) were ruptured. Cells of the endodermis and nearby stele underwent a wound reaction characterized by the deposition of brown substances in their walls.

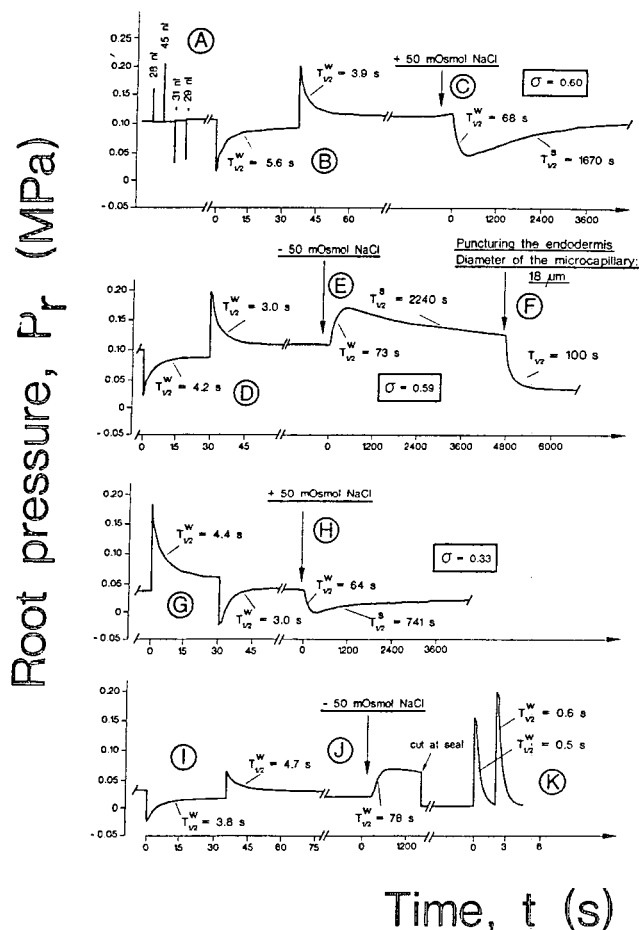


Figure 3. Time course of root pressure during a typical puncturing experiment. A, Measurements of the elasticity of the system (see "Materials and Methods"). B to E, Examples of hydrostatic and osmotic root pressure relaxations of the intact root. In B, endosmotic and exosmotic hydrostatic relaxations are given in the absence, and in D in the presence, of the osmotic solute (50 mOsmol kg⁻¹ NaCl). C and E, Biphasic osmotic pressure relaxations from which σ_{sr} and P_{sr} were calculated in addition to the osmotic Lp_r . At the conclusion of E, the root endodermis was punctured (arrow) and root pressure decreased rapidly from 0.13 to 0.04 MPa (F). From a comparison of B and D to G and I, it is evident that the hydrostatic $T_{1/2}^w$ remained the same after puncturing, i.e. Lp_r remained constant. The same is true for osmotic half-times, seen by comparing C with H. However, σ_{sr} and $T_{1/2}^s$ decreased and P_{sr} increased. At the conclusion of the experiment, the root was cut right at the seal whereupon the root pressure decreased to zero (K). Half-times decreased by an order of magnitude after the cut, proving that the root xylem in the sealing area remained open during time of the experiment (10 h). Note the different scales on the time axes.

sap would lower the concentration of solutes in the xylem until a new steady state and an osmotic equilibrium were attained. This would occur when the loss of solutes was just balanced by the active uptake of solutes (nutrients) and their delivery to the apoplast of the stele. In such a system, the absolute value of P_{ro} is a measure of the rate of active uptake of nutrients into the xylem.

To quantify the effect of changes in root transport param-

eters on the absolute value of the P_{ro} , the root may be described in terms of a pump/leak model, assuming a certain rate of active pumping of solutes (nutrients; J_{sr}^*) and some leakage caused by the permeability of the root cylinder (endodermis). The P_{sr} is a measure of the latter. Using the linear force/flow relations of irreversible thermodynamics, the radial water (J_{Vr} in m³ m⁻² s⁻¹) and solute (J_{sr} in mol m⁻² s⁻¹) flows across the root are (according to Steudle, 1989, 1992, 1993a, 1993b):

$$J_{Vr} = Lp_r (P_r - \sigma_{sr} RT [C_s^i - C_s^o]), \quad (4)$$

and

$$J_{sr} = P_{sr} (C_s^i - C_s^o) + (1 - \sigma_{sr}) \bar{C}_s J_{Vr} + J_{sr}^*, \quad (5)$$

where C_s^o and C_s^i are the solute concentrations in the medium and root xylem, respectively, $\bar{C}_s$ is the mean of both concentrations, R is gas constant, and T is temperature. By convention, $J_{sr}/J_{Vr} < 0$ denotes an uptake and $J_{sr}/J_{Vr} > 0$ indicates a loss of solutes and water from the root. For the steady state, it is valid that $J_{sr}/J_{Vr} = 0$. From Equations 4 and 5 (Steudle, 1993a, 1993b; Birner and Steudle, 1993), this yields a P_{ro} of:

$$P_{ro} = -\sigma_{sr} RT \frac{J_{sr}^*}{P_{sr}}. \quad (6)$$

According to Equation 6, P_{ro} will decrease with decreasing σ_{sr} (decreasing "passive root selectivity") and will decrease with increasing solute permeability P_{sr} (solute leakage). The equation can be used to (a) calculate the rate of active pumping (J_{sr}^*), if σ_{sr} , P_{sr} , and P_{ro} are measured (as in this paper), and to (b) calculate changes in P_{ro} in modified roots from changes in σ_{sr} and P_{sr} and compare the calculated values with measured values, assuming that J_{sr}^* remained constant. In this paper, typical values for intact roots were $P_{ro} = 0.15$ MPa, $P_{sr} = 3.8 \times 10^{-9}$ m s⁻¹, and $\sigma_{sr} = 0.64$. According to Equation 6, this results in an active $J_{sr}^* = 360 \times 10^{-9}$ mol m⁻² s⁻¹, which is similar to uptake rates measured for excised maize roots for main nutrients (e.g. 30–200 10^{-9} mol of K⁺ m⁻² s⁻¹; Anderson, 1976). On the other hand, changes of σ_{sr} from 0.64 to 0.41 and of P_{sr} from 3.8 to 10.1×10^{-9} m s⁻¹ upon puncturing (Table II) result in a calculated reduction of P_{ro} by 76% of the original or from $P_{ro} = 0.15$ MPa (on average) to 0.04 MPa. This is similar to the measured value (0.04 MPa on average; Table I), although the effect of puncturing on the absolute value of P_{ro} was rather variable. The variability could have been due to changes in J_{sr}^* and/or to variations in the area of the endodermis injured. For example, when using the same capillary tube, hitting the endodermis at right angles should have resulted in a smaller hole than a glancing blow because of the tendency of the tissue to tear for a distance along the endodermal cells in the latter case. On the other hand, pores produced by puncturing could partially close by clogging with debris or by an enlargement of the surrounding turgescient tissue. After multiple puncturing, P_{ro} declined stepwise as expected from Equation 6, because of a further decrease in σ_{sr} and an increase in P_{sr} (Fig. 4). Thus, the simple pump/leak model applied to the root explained the measured changes in root pressure reasonably well. It has been demonstrated that it may be used to calculate rates of active uptake of nutrients into roots.

Table I. Effect of puncturing the endodermis of young maize roots with a glass cannula (tip diameter: 18 to 60 μm) on stationary root pressure (P_{ro}) and hydrostatic $T_{1/2}^w$ as a measure of L_p .

Root pressures decreased considerably, but $T_{1/2}^w$ (L_p) remained unchanged, i.e. the ratio of $T_{1/2}^w$ values was not significantly different from unity (t test; $P = 0.05$). Means $\pm$ SD and ranges are given for $n = 10$ roots. The data for $T_{1/2}^w$ for individual roots are mean values and SD for $n = 6$ experiments on a given root.

Root Number	P_{ro}		P_{ro} before/ P_{ro} after	$T_{1/2}^w$		$T_{1/2}^w$ before/ $T_{1/2}^w$ after
	Before puncturing	After puncturing		Before puncturing	After puncturing	
	MPa			s		s
1	0.15	0.02	7.5	5.0 ± 0.6	3.3 ± 0.4	1.5
2	0.15	0.04	3.7	4.1 ± 1.2	2.8 ± 0.3	1.5
3	0.10	0.04	2.5	3.9 ± 0.4	3.8 ± 0.5	1.0
4	0.16	0.07	2.3	5.8 ± 0.7	5.0 ± 0.3	1.2
5	0.14	0.05	2.8	2.7 ± 0.4	3.4 ± 0.8	0.8
6	0.08	0.05	1.6	3.1 ± 0.3	3.1 ± 0.5	1.0
7	0.15	0.05	3.0	4.3 ± 0.7	4.6 ± 0.9	0.9
8	0.19	0.06	3.1	4.3 ± 0.9	4.2 ± 0.3	1.0
9	0.15	0.03	5.0	3.8 ± 1.3	2.6 ± 0.4	1.5
10	0.19	0.06	3.1	3.5 ± 0.2	2.8 ± 0.6	1.3
Mean	0.15 ± 0.03	0.05 ± 0.01	3.5 ± 1.7	4.1 ± 0.9	3.6 ± 0.8	1.2 ± 0.3
Range	0.08–0.19	0.02–0.07	1.6–7.5	2.7–5.8	2.6–5.0	0.8–1.5

Effects of Endodermal Injury on Root Hydraulic Properties ($T_{1/2}^w$ and L_p)

In both intact and modified roots, there were large differences between root hydraulic conductivities obtained from osmotic and hydrostatic experiments. This is a well-known phenomenon found for maize and other roots (Steudle et al., 1987; Cruz et al., 1992; Steudle, 1992, 1993a). It indicates substantial differences in the pathway of water movement depending on the nature of the driving force. In the case of hydrostatic pressure gradients, radial water flow is predominantly apoplasmic (i.e. circumventing protoplasts) in the root

cylinder. By contrast, there is a considerable cell-to-cell component of radial water flow in the presence of osmotic gradients. The reason for this difference, which has been observed to date in several (but not all) species, is that osmotic gradients across the apoplast constitute only a very small driving force for water, because the σ_s of the wall material for solutes is close to zero. Thus, the variable root hydraulic conductivity is a consequence of the composite nature of root transport, i.e. it is due to the fact that there are two different parallel pathways in the root contributing to the overall water transport.

Despite the large changes in the P_{ro} caused by puncturing,

Table II. Effects of endodermal puncture on parameters determined from osmotic pressure relaxations

$T_{1/2}^s$, σ_{sr} , and P_{sr} were measured using NaCl as the osmoticum (osmotic concentration: 50 mOsmol kg^{-1} of NaCl). Changes in $T_{1/2}^s$, σ_{sr} , and P_{sr} upon puncturing were significant (t test; $P = 0.05$). Individual roots (1–10) were the same as in Table I.

Root Number	$T_{1/2}^s$		$T_{1/2}^s$ before/ $T_{1/2}^s$ after	σ_{sr}		σ_{sr} before/ σ_{sr} after	P_{sr}	
	Before puncturing	After puncturing		Before puncturing	After puncturing		Before puncturing	After puncturing
	s						$(\text{m s}^{-1}) 10^9$	
1	1500	1910	0.8	0.73	0.38	1.9	3.2	2.5
2	1720	385	4.5	0.77	0.59	1.3	2.7	12.3
3	1670	740	2.3	0.59	0.33	1.8	3.9	8.9
4	1060	520	2.0	0.76	0.44	1.7	4.8	9.9
5	985	1670	0.6	0.74	0.27	2.7	5.8	3.4
6	3900	2460	1.6	0.41	0.22	1.9	1.2	1.9
7	890	780	1.1	0.92	0.45	2.0	4.5	5.2
8	1630	225	7.2	0.37	0.28	1.3	3.3	24.1
9	1520	350	4.3	0.70	0.67	1.0	3.1	13.5
10	1030	285	3.6	0.40	0.43	0.9	5.2	19.0
Mean	1590 ± 870	930 ± 790	2.8 ± 2.1	0.64 ± 0.19	0.41 ± 0.14	1.7 ± 0.5	3.8 ± 1.4	10.1 ± 7.4
Range	890–3900	225–2460	0.6–7.2	0.37–0.92	0.22–0.67	0.9–2.7	1.2–5.8	1.9–24.1

Table III. Summary of the effects of puncturing maize roots on P_{ro} , Lp_r , σ_{sr} , and P_{sr}

Note that the average hydrostatic Lp_r was about 10-fold larger than the osmotic. The values are means $\pm$ SD for the same 10 roots as in Tables I and II. Changes in P_{ro} , σ_{sr} , and P_{sr} were significant (see Tables I and II), but there were no changes in either the hydrostatic or osmotic Lp_r after puncturing.

	Before Puncturing	After Puncturing
P_{ro} (MPa)		
Mean	0.15 ± 0.03	0.05 ± 0.01
Range	0.08–0.19	0.02–0.07
Hydrostatic $Lp_r \times 10^7$ ($\text{m s}^{-1} \text{MPa}^{-1}$)		
Mean	2.7 ± 1.3	2.8 ± 1.6
Range	0.8–6.3	0.7–6.7
Osmotic $Lp_r \times 10^8$ ($\text{m s}^{-1} \text{MPa}^{-1}$)		
Mean	2.2 ± 2.1	2.5 ± 2.5
Range	0.3–6.4	0.7–7.5
σ_{sr}		
Mean	0.64 ± 0.19	0.41 ± 0.14
Range	0.37–0.92	0.22–0.67
$P_{sr} \times 10^9$ (m s^{-1})		
Mean	3.8 ± 1.4	10.1 ± 7.4
Range	1.2–5.8	1.9–24.1

hydraulic conductivity measured either by hydrostatic or osmotic experiments was not affected. This was so, even though, according to Poiseuille's law, a hole with a diameter of $20 \mu\text{m}$ and a length of $400 \mu\text{m}$ (i.e. across the entire root cylinder up to the xylem) would exhibit the very high hydraulic conductance of $0.98 \times 10^{-8} \text{ m}^3 \text{ s}^{-1} \text{MPa}^{-1}$ in hydrostatic experiments (viscosity of water at 20°C : 10^{-3} Pa s). Expressing this conductance in relation to the total A_r results in an Lp_r as large as $3.8 \times 10^{-5} \text{ m s}^{-1} \text{MPa}^{-1}$, i.e. a value that is larger by 2 orders of magnitude than that found for both intact and punctured roots (hydrostatic $Lp_r = 2.7 \times 10^{-7} \text{ m s}^{-1} \text{MPa}^{-1}$). The reason for this discrepancy may be that the holes (pores) made were not entirely across the stelar tissue up to the vessels and across the vessel walls. Damaging the vessels resulted in irreversible damage to the root, probably because the outflow of solutes from the xylem could no longer be compensated for by active pumping. Thus, the situation in the experiments could be different from that in the calculation given above.

The lack of any change in the osmotic Lp_r is to be expected. In contrast to hydrostatic gradients, osmotic gradients across the pore could not have caused water flow because there was no selective membrane on this path. The lack of change in the hydrostatic and osmotic Lp_r to puncturing shows that the radial hydraulic resistance of the root was rather evenly distributed over all of its tissues. This idea differs considerably from the commonly held view of the endodermis being the main hydraulic resistance in roots. It is supported by recent measurements of Peterson et al. (1993), who removed some of the root cortex of young maize roots by scraping and also punctured roots with thin glass cannulae. In both cases the hydrostatic Lp_r remained unchanged despite reductions in root pressure that occurred during puncturing.

Location of Barriers to Solute Flow: Effect of Puncturing on Root P_{sr} and σ_{sr}

The effect of puncturing on the root P_{sr} and σ_{sr} was substantial, indicating that the endodermis was the main barrier for solute transport. Thus, the contribution of the apoplastic solute transport in regions lacking a Casparian band (root initials, tip route) to overall solute transport is potentially important, although the mobility of solutes in these pathways is known to be restricted (Peterson et al., 1981; Enstone and

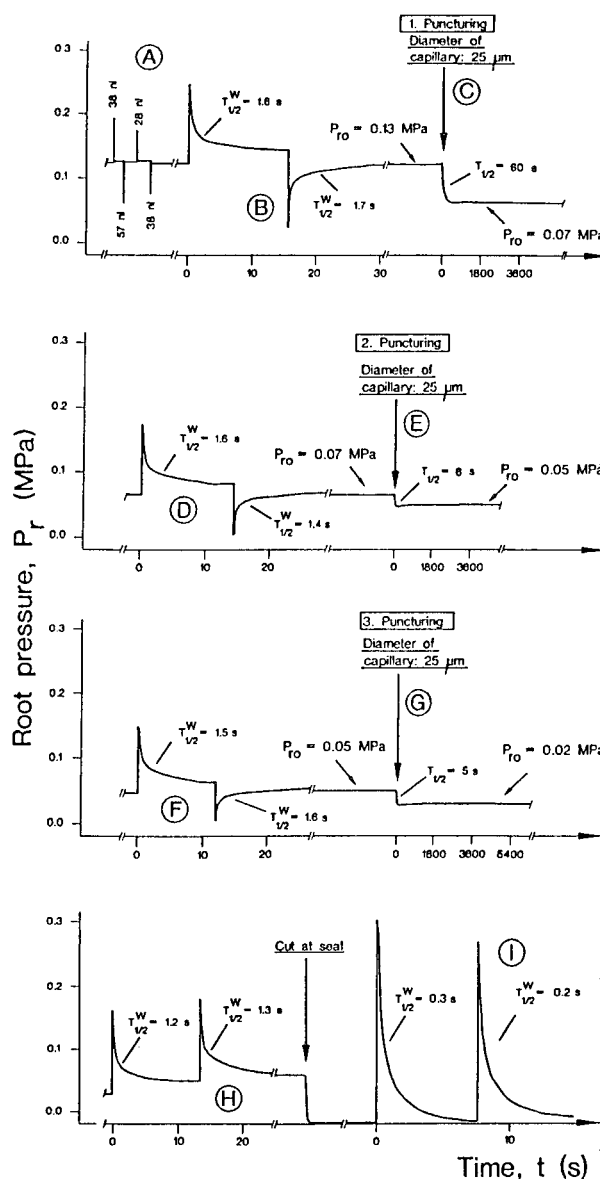


Figure 4. Effect of multiple puncturing of the endodermis on root pressure. Hydrostatic relaxations; B, D, F, and H are as described in Figure 3. The first puncturing (C) resulted in a decrease of root pressure from 0.13 to 0.07 MPa, and the second (E) resulted in a decrease from 0.07 to 0.05 MPa. The third hole (G) caused a further pressure decrease from 0.05 to 0.03 MPa, i.e. to a rather low value. Note that multiple puncturing did not change $T_{1/2}^W$ (and therefore Lp_r) in hydrostatic pressure relaxations.

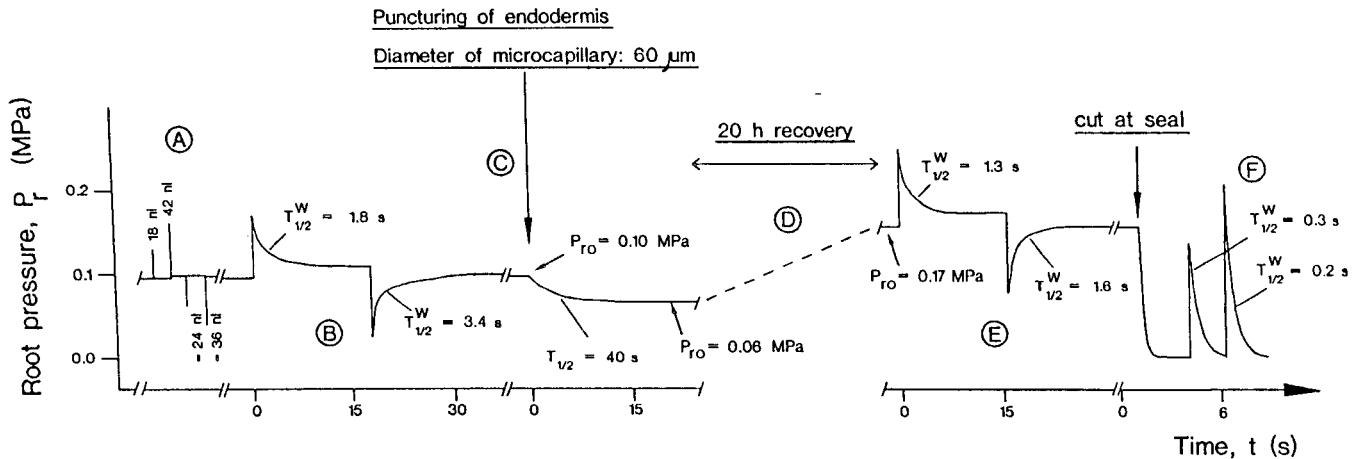


Figure 5. Typical experiment demonstrating the recovery of root pressure in punctured roots. The different regions (A to C and E and F) are as explained in Figure 3. D, Recovery took about 20 h for this root. During this time the root regained its ability to accumulate ions in the stele (which includes the xylem), thus increasing its root pressure. In this particular root, the root pressure after recovery was even larger than the original.

Peterson, 1992). The only apoplastic pathway from the cortex to the main root stele that is open to molecules the size of fluorescent dyes exists at the flanks of emerged lateral roots (Peterson et al., 1981).

Solute transport can be analyzed in detail by using the experimental data (permeability coefficients and areas available for solute transport in different pathways). Of interest is the contribution of different pathways to the overall solute transport in the roots and the solute mobility in these pathways. For this evaluation the solute transport across the Casparian band itself has to be taken into account as well, because this structure occupied an area as large as 12% of the total surface area of the endodermis. Two extreme cases are of interest: (a) the case in which the Casparian band is completely impermeable to solutes (the commonly accepted

view) and (b) the case in which the transport across the Casparian band and apoplastic bypasses (tip and lateral roots combined) contribute equally to the overall solute permeability. In the latter case, the Casparian band would be extremely leaky. Transmembrane solute transport across endodermal cells may be considered negligible in this context to a first approximation (but see below).

(a) If the Casparian band were completely impermeable to solutes, in this case NaCl, a permeability coefficient for the transport across secondary root initials and the main root tip may be estimated by first referring the root P_{sr} of $3.8 \times 10^{-9} \text{ m s}^{-1}$ (Table II) to the surface area of the endodermis (as the limiting barrier) instead of that of the root surface. Because the surface area of the endodermis was about half that of the root, this yields a $P_{sr} = 7.6 \times 10^{-9} \text{ m s}^{-1}$ for the endodermis.

Table IV. Effects of recovery on P_{ro} and T_{v2}^w in punctured roots

Recovery took from 1.5 to 20 h, after which a root pressure equal to or even higher than the original was attained. Hydrostatic T_{v2}^w (Lp_r) did not change during recovery.

Root Number	Time Required for Recovery	P_{ro}			T_{v2}^w	
		Before puncturing	After puncturing	After recovery	Before puncturing	After recovery
	<i>h</i>		<i>MPa</i>			<i>s</i>
11	7.5	0.17	0.05	0.17	4.7 ± 0.7	9.6 ± 0.2
12	12	0.17	0.07	0.15	4.0 ± 0.5	5.2 ± 0.6
13	17	0.11	0.04	0.20	7.7 ± 0.5	5.9 ± 1.4
14	13	0.17	0.05	0.22	3.6 ± 0.3	2.7 ± 0.5
15	14	0.16	0.12	0.27	5.3 ± 0.4	4.6 ± 0.6
16	10	0.09	0.02	0.05	4.5 ± 0.7	6.2 ± 1.2
17	16	0.08	0.03	0.02	3.2 ± 0.1	7.2 ± 3.1
18	11	0.12	0.04	0.19	3.6 ± 0.7	2.9 ± 0.5
19	1.5	0.07	0.04	0.08	5.7 ± 1.1	6.8 ± 1.0
20	20	0.10	0.06	0.17	2.2 ± 0.6	1.6 ± 0.2
Mean	12.2 ± 5.2	0.12 ± 0.04	0.05 ± 0.03	0.15 ± 0.08	4.5 ± 1.5	5.3 ± 2.4
Range	1.5–20	0.07–0.17	0.02–0.12	0.02–0.27	2.2–7.7	1.6–9.6

If the total flow is assumed to go through the walls in areas lacking a Casparian band ($8 \times 10^{-2}\%$ of the surface area of the endodermis), we get a $P_{sr} = 7.6/0.08 \times 10^{-7} = 9.5 \times 10^{-6} \text{ m s}^{-1}$ for that apoplastic path. Assuming a homogeneous path of length of $12 \text{ }\mu\text{m}$ (the thickness of the endodermis), an apoplastic diffusion coefficient (D_s) for NaCl of $D_s = 9.5 \times 1.2 \times 10^{-11} = 1.1 \times 10^{-10} \text{ m}^2 \text{ s}^{-1}$ is estimated. This value is within an order of magnitude of those given in the literature for Na^+ and Cl^- ions in walls of barley root tissue (3×10^{-11} to $10 \times 10^{-11} \text{ m}^2 \text{ s}^{-1}$; for refs., see Walker and Pitman, 1976). It is considerably smaller than that of NaCl in bulk aqueous solution ($1.5 \times 10^{-9} \text{ m}^2 \text{ s}^{-1}$ at 25°C).

(b) If, on the other hand, the transport across the Casparian band and that across apoplastic bypasses contributed equally to the overall transport ("leaky Casparian band"), this would mean that the permeability of the Casparian band would be smaller by a factor of $12/0.080 = 150$ than that along apoplast lacking a hydrophobic barrier. Using this value and considering a thickness of the Casparian band of $2 \text{ }\mu\text{m}$ yields a $P_{sr} = 3.2 \times 10^{-8} \text{ m s}^{-1}$ and a $D_s = 6.4 \times 10^{-14} \text{ m}^2 \text{ s}^{-1}$ for the Casparian band. This value is smaller by 3 orders of magnitude than the D_s for NaCl in the apoplast lacking a hydrophobic barrier (see above), despite the fact that the calculation assumed a leakage of as large as 50% across the Casparian band. It is clear that, for a proper function of the Casparian band as an effective barrier for solutes, the D_s would have to be considerably smaller. Thus, it could become comparable to D_s values of univalent salts in cuticles (1×10^{-15} to $3 \times 10^{-17} \text{ m}^2 \text{ s}^{-1}$ for KCl; Tyree et al., 1992). The calculations show that, if the Casparian band were almost impermeable, its D_s would have to be even less than that of a waxy layer like the cuticle. This is unlikely to be the case because suberin is less hydrophobic than wax. Therefore, it is reasonable to assume that Casparian bands would allow some bypass of solutes and water even in a mature endodermis. Because the bypass would be small, it would not really affect the function of the endodermis a "filter for solutes." From root pressure probe data (Steudle, 1989, 1992, 1993a, 1993b) and also from other work (Hanson et al., 1985; Yeo et al., 1987; Radin and Matthews, 1989; Behrmann and Heyser, 1993), there is experimental evidence of apoplastic bypasses in the endodermis, but more data are undoubtedly required for proof.

Although the membrane permeability of the solute (NaCl) was not directly measured in this paper, we may estimate the possible contribution of the membrane-bound component (endodermal protoplasts) to solute transport from the overall permeability and from the morphometric data. For example, if the components across the apoplast lacking a Casparian band (0.08% of total surface area of endodermis), across the Casparian band itself (12%), and the endodermal cells (88%) each contributed equally to the overall transport (i.e. each by one-third), this would result in a permeability coefficient (P_s) of the endodermal cell membrane of $P_s = 5.8 \times 10^{-9} \text{ m s}^{-1}$. This value is somewhat higher than those given in the literature for P_s values of root cell membranes to Na^+ and Cl^- (1×10^{-9} to $3 \times 10^{-11} \text{ m s}^{-1}$; Pitman, 1976). Hence, the contribution of the cellular path may be smaller than one-third. The comparison of the data shows that the membrane-

bound path across the endodermis cannot be neglected in a rigorous evaluation, at least for an endodermis lacking suberin lamellae.

Recovery from Puncturing

Repair of punctured roots appeared to be a process that usually started 0.5 to 1 h after wounding, but complete recovery took up to 20 h. This indicates that, following an injury, roots soon improved their selectivity and their ability to accumulate nutrient ions in the stele. The rapid closure of injuries in the endodermis or stele may be favored by the lack of air spaces that would otherwise have to be filled with hydrophobic material (Peterson et al., 1993). However, the osmotic experiments attempted on recovered roots showed that they represented rather unstable osmometers, tending to leak and lose root pressure when subjected to osmotic stress. For this reason, no satisfactory osmotic experiments could be performed on recovered roots. Perhaps by improving the conditions during recovery, more stable roots could be produced in the future. This would be a worthwhile achievement because punctured roots could be "filled" with different solutes, dyes, inhibitors, hormones, and the like while the pore is open. After the pore closes, different types of modified roots would thus be obtained to study transport and other root functions.

Composite Transport Model

At first glance, it is astonishing that after making a hole as large as $60 \text{ }\mu\text{m}$ in the endodermis the roots maintained a positive root pressure and exhibited some passive selectivity, i.e. $\sigma_{sr} > 0$. The solute permeability of the modified roots was still rather low, and the osmotic responses after puncturing were basically the same as those of intact roots, i.e. they were typical for an osmometer in the presence of a permeating solute as measured for roots or plant cells and derived from irreversible thermodynamics. Similarities in osmotic responses of intact and modified roots may be explained by an extended osmometer model of the root, which is shown schematically in Figure 6. The root can be represented by an osmotic system having a composite osmotic barrier consisting of the cell-to-cell path (superscript "b"), which is an almost semipermeable barrier because of the presence of numerous membranes in series, and the nonselective (porous) wall path that should leak solutes (superscript "a"). The function of the Casparian band (Fig. 6A) is to minimize this leakage. Although there may be some leakage present in the system due to (a) a not completely tight Casparian band (Fig. 6A), (b) apoplastic pathways lacking Casparian bands (Fig. 6B), or (c) an artificial pore or injury (as in this paper; Fig. 6C), the root will respond as an osmometer because of the existence of the cell-to-cell path. If the solute concentration in the xylem is larger than that in the medium, water will be sucked into the xylem along the cell-to-cell path and root pressure will build up. However, this will immediately induce a back-flow of water (solution) across the nonselective apoplast or,

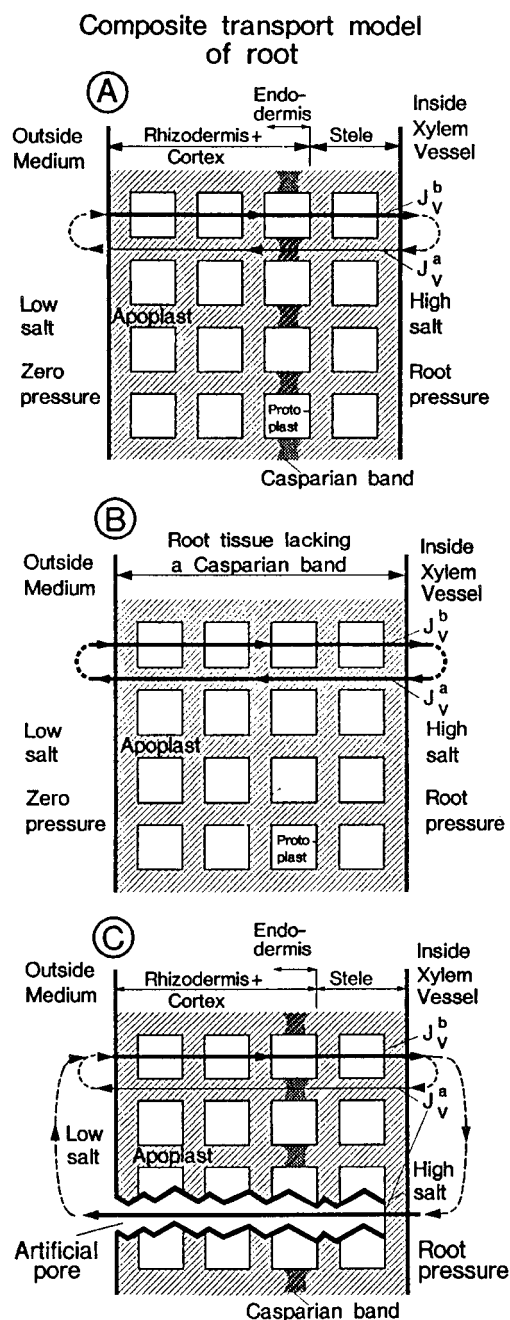


Figure 6. Diagrammatic scheme of composite transport in a plant root. A and B denote transport across the intact system (root tissue with or without an endodermis, respectively), and C depicts the situation in a punctured root. For the sake of simplicity, rhizodermis and cortex (including the endodermis) are represented by three cell layers and the stelar tissues by a single layer. In C, punctured roots vessels were not hit by the microcapillary tube. Different routes of radial flow are denoted, i.e. the flow across the apoplast or an artificial hole (superscript "a") and the cell-to-cell path (superscript "b"). The figure indicates that there will be an osmotic water flow (J_v^b) directed from the root medium into the xylem (stele) as long as the osmotic concentration in the xylem is larger than that in the medium. The σ_{sr} along this path should be high ($\sigma_s^b \approx 1$). However, as soon as a root pressure builds up, there will be a backflow of water (xylem solution) across the nonselective apo-

in the modified root, across the artificial pore. Because there is some active pumping of nutrients, a steady root pressure will be established when the rate of the osmotic water uptake just balances the hydraulic extrusion. In the steady state, the opposing water flows will cancel, and some kind of a circulation flow of water across the entire barrier will be established (Fig. 6). It is clear that, because of the backflow, the root osmometer will not behave like an ideal (semipermeable) osmotic cell. In osmotic equilibrium, the hydrostatic pressure difference set up will be smaller than the osmotic pressure difference between the xylem sap and medium. In quantitative terms, this is expressed by $\sigma_{sr} < 1$, and it will hold that $P_{ro} = \sigma_{sr} \cdot \Delta\pi_s$. The rate of backflow will increase with increasing size and number of holes (leaks), and P_{ro} and σ_{sr} will decrease further as demonstrated in this paper. On the other hand, the backflow will be reduced by a decreasing permeability of the Casparian band during root development, and it is clear that σ_{sr} will increase with a decreasing hydraulic conductance of the endodermal or other apoplastic path.

Composite systems such as that depicted for the root in Figure 6 have been readily treated in the theory of the thermodynamics of irreversible processes as applied to membranes composed of arrays with different transport properties (Kedem and Katchalsky, 1963; Steudle, 1989, 1992, 1993a, 1993b). The theory shows that for a system composed of two parallel pathways we get for the overall σ_{sr} of the barrier (root):

$$\sigma_{sr} = \sigma_s^a \frac{\gamma^a Lp^a}{Lp_r} + \sigma_s^b \frac{\gamma^b Lp^b}{Lp_r}. \quad (7)$$

Here the same notation is used as in Figure 6, i.e. the superscript "a" for transport across the hole or apoplast and the superscript "b" for the cell-to-cell path. The γ^a and γ^b denote the fractional cross-sectional areas of the two different pathways ($\gamma^a + \gamma^b = 1$) and Lp^a and Lp^b represent their hydraulic conductivities. The σ_s of the two different arrays are denoted by σ_s^a , and σ_s^b . It can be seen from Equation 7 that the overall σ_{sr} is a weighted mean of the individual σ_s values and that the different components of the barrier contribute to the overall σ_{sr} according to their hydraulic conductance ($\gamma^a Lp^a$ and $\gamma^b Lp^b$, respectively), whereby it holds that $Lp_r = \gamma^a Lp^a + \gamma^b Lp^b$. Hence, the contribution of the very low σ_s of the apoplastic or pore path lacking a Casparian band ($\sigma_s^a \approx 0$) to the overall σ_{sr} could be high because of the high hydraulic conductance of the apoplast or pore, although the area available for apoplastic transport is much smaller than that along the cell-to-cell path.

plasmic paths ($\sigma_s^a \approx 0$) in B, C, and, perhaps, also in A if the mature Casparian band is somewhat permeable to water. When the opposing water flows exactly cancel each other, root pressure will hold steady and will be smaller than expected on the basis of the difference in osmotic pressure between the xylem sap and the medium, i.e. σ_{sr} will be smaller than unity. It can be seen from the model that, according to the composite structure of the entire barrier, a circulation flow of water will be set up across the root at zero or low net water flow (J_{vr}).

For the same system composed of two parallel pathways, the overall P_{sr} will be given by (Kedem and Katchalsky, 1963; see also House, 1974):

$$P_{sr} = \gamma^a P_s^a + \gamma^b P_s^b + (\sigma_s^b - \sigma_s^a)^2 RT \frac{Lp^a Lp^b}{Lp_r} \bar{C}_s. \quad (8)$$

Again, the superscripts "a" and "b" denote properties of individual pathways as above. The first two terms on the right side of Equation 8 represent the contributions of the diffusional components of the individual pathways to the overall solute flow. The additional component (third term on the right side), which tends to increase P_{sr} , results from the solvent drag component of solute flow (see second term on the right side of Equation 5; for details of the theory, see refs. given). Thus, an increase of P_{sr} caused by puncturing the endodermis was due both to the creation of an additional diffusional path for solutes ($\gamma^a P_s^a$) and to a solvent-drag component. For the osmotic experiments with punctured roots, it is hard to judge what the relative contributions of the different components (solvent drag versus diffusional) would be. However, owing to the rapid adjustment of root pressure to the new stationary value after puncturing (i.e. to the large water flow) and to the rather long half-times of solute flow found before and after puncturing, the effect of solvent drag should be considerable. Experimentally, it would be difficult to separate the different components given in Equation 8. The same applies to direct measurements of the inwardly and outwardly directed water flows in the system.

The composite transport model predicts some variability in root hydraulic resistances, i.e. a decrease of the hydraulic resistance with increasing water flow across the root. This effect has been known for a long time, but no satisfactory explanation has been given for it (Fiscus, 1975; Weatherley, 1982; Passioura, 1988; Katou and Taura, 1989). The composite model could resolve the problem. The model predicts that at low rates of water flow, when the osmotic component in the driving force would be substantial and water flows in the root would, in part, cancel, the resistance should be high (Lp_r low). This should be the case at zero or low rates of transpiration. At higher flow rates (higher rates of transpiration), tensions in the root xylem (i.e., the hydrostatic component) will dominate the overall driving force for water. This will result in a water flow along the two pathways in the same direction and in a lower hydraulic resistance of the root (Lp_r high).

The composite transport approach integrates quite different and important properties of roots in a single model. It has a sound theoretical and anatomical basis. It may be extended to roots exhibiting zones of various developmental stages by treating these zones as arrays of different transport properties arranged in parallel, and to entire root systems (Steudle, 1989, 1992, 1993a, 1993b).

ACKNOWLEDGMENTS

The authors thank Marcus Rüdinger for his help in performing some of the puncturing experiments and Burkhard Stumpf for his

expert technical assistance. The help of Ms. Daryl Enstone in producing Figure 2, A, C, and D, is also gratefully acknowledged.

Received April 14, 1993; accepted June 25, 1993.

Copyright Clearance Center: 0032-0889/93/103/0335/15.

LITERATURE CITED

- Anderson WP (1976) Transport through roots. In U Lüttge, MG Pitman, eds, *Encyclopedia of Plant Physiology*, Vol 2, Part B. Springer-Verlag, Heidelberg, Germany, pp 129–156
- Azaiz H, Steudle E (1991) Effects of salinity on water transport of excised maize (*Zea mays* L.) roots. *Plant Physiol* 97: 1136–1145
- Behrmann P, Heyser W (1993) Apoplastic transport through the fungal sheath of *Pinus sylvestris*/Suillus bovinus ectomycorrhizae. *Bot Acta* 105: 427–434
- Bell JK, McCully ME (1970) A histological study of lateral root initiation and development in *Zea mays*. *Protoplasma* 70: 179–205
- Birner TP, Steudle E (1993) Effects of anaerobic conditions on water and solute relations and active transport in roots of maize (*Zea mays* L.). *Planta* 190: 474–483
- Brundrett MC, Enstone DE, Peterson CA (1988) A berberine-aniline blue fluorescent staining procedure for suberin, lignin and callose in plant tissue. *Protoplasma* 146: 133–142
- Cruz RT, Jordan WR, Drew MC (1992) Structural changes and associated reduction of hydraulic conductance in roots of *Sorghum bicolor* L. following exposure to water deficit. *Plant Physiol* 99: 203–212
- Enstone DE, Peterson CA (1992) The apoplastic permeability of root apices. *Can J Bot* 70: 1502–1512
- Fiscus EL (1975) The interaction between osmotic- and pressure-induced water flow in plant roots. *Plant Physiol* 55: 917–922
- Frensch J, Stelzer R, Steudle E (1992) NaCl uptake in roots of *Zea mays* seedlings: comparison of root pressure probe and EDX data. *Ann Bot* 70: 543–550
- Frensch J, Steudle E (1989) Axial and radial hydraulic resistance to roots of maize (*Zea mays* L.). *Plant Physiol* 91: 719–726
- Hackett C, Stewart HE (1969) A method for determining the position and size of lateral primordia in the axes of roots without sectioning. *Ann Bot* 33: 679–682
- Hanson PJ, Sucoff EI, Markhart AH (1985) Quantifying apoplastic flux through red pine root systems using trisodium 3-hydroxy-5,8,10-pyrenetrisulfonate. *Plant Physiol* 77: 21–24
- House CR (1974) *Water Transport in Cells and Tissues*. Edward Arnold, London
- Karas I, McCully ME (1973) Further studies of the histology of lateral root development in *Zea mays*. *Protoplasma* 77: 243–269
- Katou K, Taura T (1989) Mechanism of pressure-induced water flow across plant roots. *Protoplasma* 150: 124–130
- Kedem O, Katchalsky A (1963) Permeability of composite membranes, Part 2: Parallel elements. *Trans Far Soc* 59: 1931–1940
- Melchior W, Steudle E (1993) Water transport in onion (*Allium cepa* L.) roots. Changes of axial and radial hydraulic conductivities during root development. *Plant Physiol* 101: 1305–1315
- Passioura JB (1988) Water transport in and to roots. *Annu Rev Plant Physiol Plant Mol Biol* 39: 245–265
- Peterson CA, Emanuel ME, Humphreys GB (1981) Pathway of movement of apoplastic fluorescent tracers through the endodermis at the site of secondary root formation in corn (*Zea mays*) and broad bean (*Vicia faba*). *Can J Bot* 59: 618–625
- Peterson CA, Murrmann M, Steudle E (1993) Location of the major barriers to water and ion movement in young roots of *Zea mays* L. *Planta* 190: 127–136
- Peterson CA, Steudle E (1993) Lateral hydraulic conductivity of early metaxylem vessels in *Zea mays* L. roots. *Planta* 189: 288–297
- Pitman MG (1976) Ion uptake by plant roots. In U Lüttge, MG Pitman, eds, *Encyclopedia of Plant Physiology*, Vol 2B. Springer-Verlag, Berlin, pp 95–128
- Radin JW, Matthews MA (1989) Water transport properties of cells in the root cortex of nitrogen- and phosphorus-deficient cotton seedlings. *Plant Physiol* 89: 264–268
- Steudle E (1989) Water flow in plants and its coupling to other processes: an overview. *Methods Enzymol* 174: 183–225

- Steudle E** (1992) The biophysics of plant water: compartmentation, coupling with metabolic processes, and water flow in plant roots. In GN Somero, CB Osmond, CL Bolis, eds, *Water and Life: Comparative Analysis of Water Relationships at the Organismic, Cellular, and Molecular Levels*. Springer-Verlag, Berlin, pp 173–204
- Steudle E** (1993a) Pressure probe techniques: application to studies of water and solute relations at the cell, tissue and organ levels. In JAC Smith, H Griffiths, eds, *Water Deficits: Plant Responses from Cell to Community*. Bios Scientific, Oxford, UK (in press)
- Steudle E** (1993b) The regulation of plant water at the cell, tissue and organ level: role of active processes and of compartmentation. In E-D Schulze, ed, *Flux Control in Biological Systems: from the Enzyme to Ecosystem Level*. Academic Press, New York (in press)
- Steudle E, Brinckmann E** (1989) The osmometer model of the root: water and solute relations of roots of *Phaseolus coccineus*. *Bot Acta* **102**: 85–95
- Steudle E, Frensch J** (1989) Osmotic responses of maize roots. Water and solute relations. *Planta* **177**: 281–295
- Steudle E, Jeschke WD** (1983) Water transport in barley roots. *Planta* **158**: 237–248
- Steudle E, Oren R, Schulze ED** (1987) Water transport in maize roots. Measurement of hydraulic conductivity, solute permeability, and of reflection coefficients of excised roots using the root pressure probe. *Plant Physiol* **84**: 1220–1232
- Tyree MT, Wescott CR, Tabor CA, Morse AD** (1992) Diffusion and electric mobility of KCl within isolated cuticles of *Citrus aurantium*. *Plant Physiol* **99**: 1057–1061
- Walker NA, Pitman MG** (1976) Measurement of fluxes across membranes. In U Lüttge, MG Pitman, eds, *Encyclopedia of Plant Physiology*, Vol 2A. Springer-Verlag, Berlin, pp 93–126
- Weatherley PJ** (1982) Uptake and flow in roots. In OL Lange, CB Osmond, H Ziegler, eds, *Encyclopedia of Plant Physiology*, Vol 12B: Physiological Plant Ecology. Springer-Verlag, Heidelberg, Germany, pp 79–109
- Yeo AR, Yeo ME, Flowers TJ** (1987) The contribution of an apoplastic pathway to sodium uptake by rice roots in saline conditions. *J Exp Bot* **38**: 1141–1153
- Zhu GL, Steudle E** (1991) Water transport across maize roots. Simultaneous measurement of flows at the cell and root level by double pressure probe technique. *Plant Physiol* **95**: 305–315